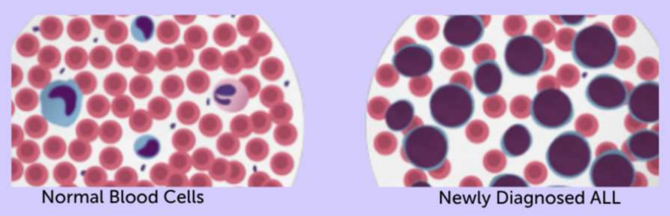
INVESTGIATING HUMAN LEUKEMIA WHEN EXPOSED ZNPP TREATMENTS TO INCREASE OXIDATIVE STRESS

BACKGROUND

Acute Lymphoblastic Leukemia

Acute Lymphoblastic Leukemia: rapidly progressive blood cancer caused by overproduction of immature lymphoblasts in bone marrow and bloodstream



Annual Incidence

386,813

New ALL prevalent cases the by year 2021. (PudMed Central)

ALL represents 20% of all leukemia in the world and is the common childhood blood cancer.

Peak Incidence: Age 2-5 years (children); secondary peak in adults after age 50. Without treatment: Median survival 2-3 months. Modern therapy: 5-year now 90% in children and 40-50% in adults (NCCN Gudelines 2024, SEER Database)

Proposed Treatment:

- Zinc protoporphyrin (ZnPP) is proposed as a targeted, alternative therapeutic agent for ALL. (Metz et al., 2010)
- The HO-1 pathways produces **anxtioxidants** to decreased **oxidative stress** levels. (Pae & Chung, 2009)
- ZnPP **inhibits** the **HO-1 pathway** to increase Oxdiatve Stress in the cells. (Nowis et al., 2008)
- Why this treatment: By **blocking** the **HO-1 enzyme** in cancer cells to chemotherapy and evade cell death, allowing cell apoptosis. (Mei et al., 2026)

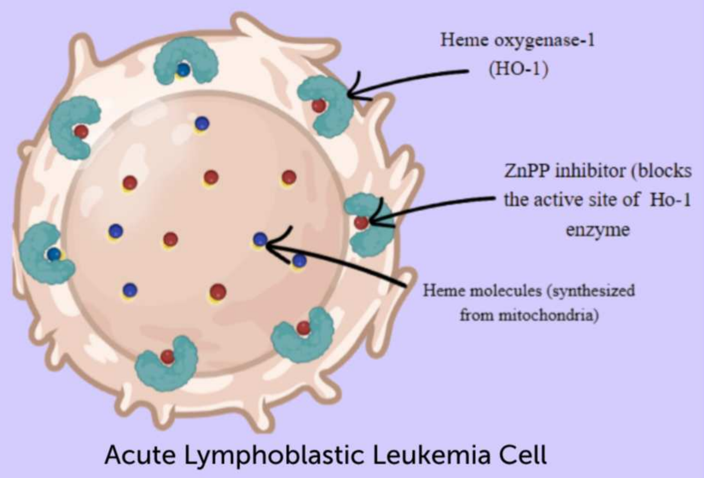


Figure 1: Background research on Acute Lymphoblastic Leukemia Cells.

PURPOSE

- **This study investigates** the effects of ZnPP on inhibiting the HO-1 pathway in ALL cells.
- **Mechanism**: Blocking HO-1 raises oxidative stress within the cell, damaging it and triggering apoptosis
- **Goal**: Determine whether ZnPP has potential as a future component of ALL treatment

HYPOTHESIS & VARIABLES

Hypothesis: If Jurkat E6-1 cells are treated with increasing concentrations of ZnPP, then cell viability will decrease, because HO-1 inhibition elevates oxidative stress, leading to greater cellular damage and apoptosis.

IV: Concentration of ZnPP
(20μM, 10μM, 5μM, 2.5μM)

DV: Cell viability measured by Cell TiterGlo Assay

Negative Control: Untreated Jurkat E6-1 cells without ZnPP

METHODOLOGY

IV Concentration treatments:

2.5 µm ZnPP stock
5 µm ZnPP stock
10 µm ZnPP stock
20 µm ZnPP stock

Negative Control:
(Media + Cells)

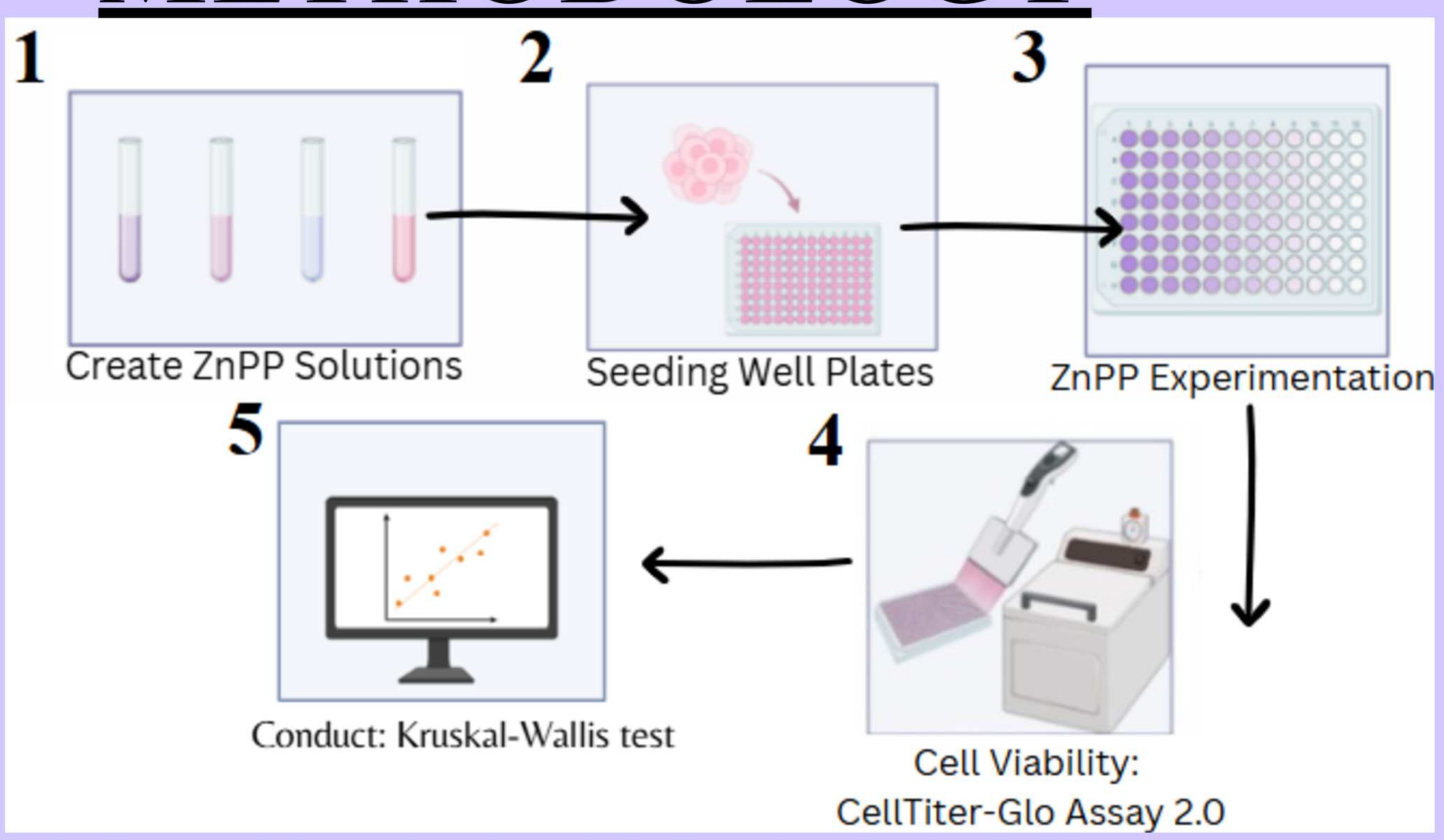


Figure 2. Cell TiterGlo assay measures ATP of cells to track the number of metabolically active cells after ZnPP treatment

DATA & RESULTS

CELL VIABILITY ASSAY

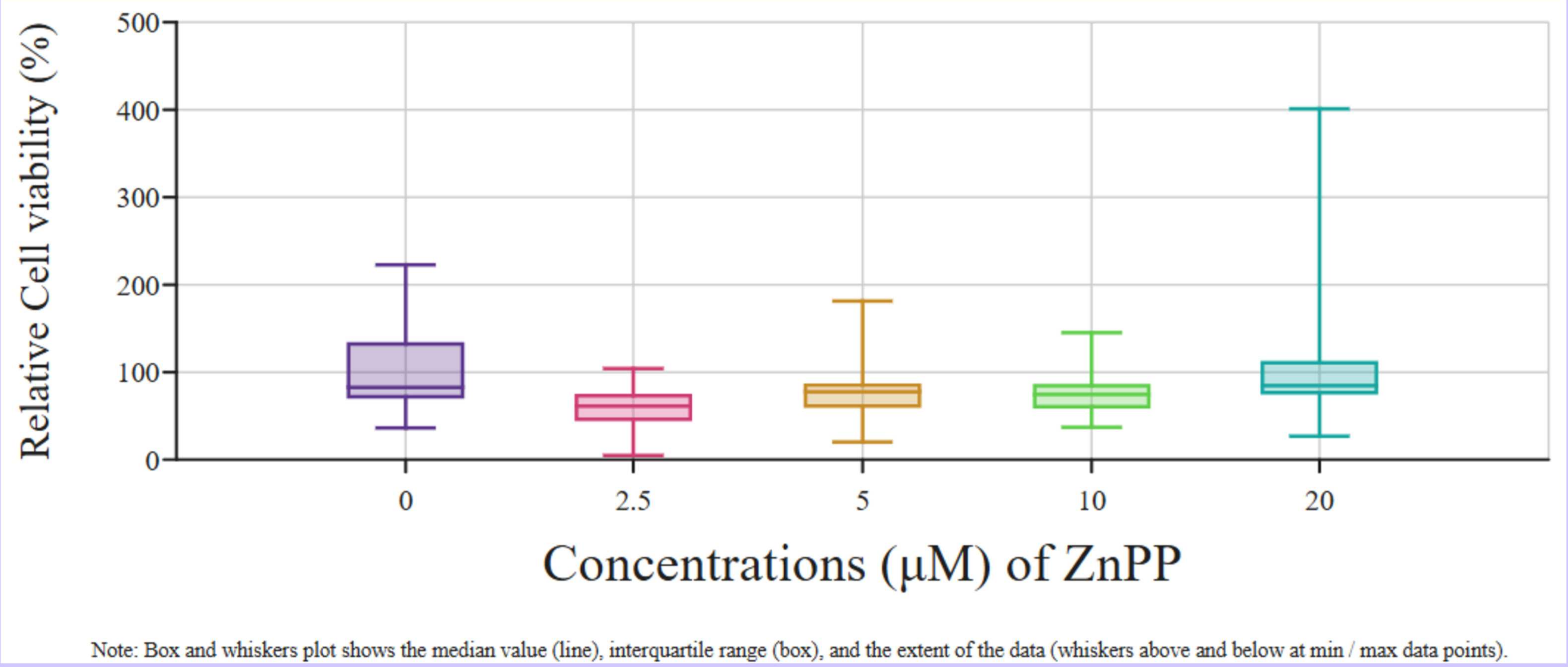


Figure 3: Cell TiterGlo Assay cell Viability for each ZnPP concentration with a Kruskal Wallis test

KRUSKAL WALLIS TEST

Figure 4: **the groups highlighted in blue show extreme strong confidence that the groups are different while all groups highlighted in red are not different.**

Pair (µM)	P-value	Kruskal Wallis P-Value
0 - 5	<0.01	<0.01 (A P-Value of <0.01 means extremely strong confidence that the groups are different.)
0 - 10	0.61	
0 - 20	0.18	
0 - 2.5	1	
5 - 10	0.13	
5 - 20	0.36	
5 - 2.5	<0.01	
10 - 20	1	
10 - 2.5	0.51	
20 - 2.5	0.17	

Figure 4. The Kruskal Wallis Test Table from Cell TiterGlo Assay Results

DISCUSSION

Findings

- ZnPP killed the most cells at 2.5 µM while at higher concentrations (5, 10, 20 µM), fewer cells died. This unexpected pattern suggests ZnPP works differently at different doses and needs further study.
- Kruskal-Wallis test confirmed statistically significant differences among ZnPP concentration groups.
- $H = 29.11$, $p < 0.01$)
- Post-hoc analysis revealed significant differences between 0–5 µM concentrations, indicating a possible non-linear relationship rather than a linear dose-response.

Implications

- ZnPP successfully influences Jurkat cell viability through oxidative stress mechanisms.
- Concentration-dependent effects suggest optimal therapeutic dosing exists within tested range.
- Variability within treatment groups suggest repeating the experiment to confirm the reliability of ZnPP's effects.

Limitations

- CellTiter-Glo measures metabolic activity, not direct oxidative damage confirmation
- Lack of linear trend suggests additional cellular mechanisms may be at play
- Higher concentrations may not reflect actual cell numbers, since metabolic changes can affect luminescence values independent of cell death.
- CellTiter-Glo assay suggests concentration-dependent effects on viability cannot be distinguished from other metabolic changes vs. true cell death.

FUTURE WORK

- TBARS assay will be conducted to directly measure oxidative damage from each concentrations.
- Further trials on Cell TiterGlo Assay with vehicle control

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